

CH 1.1: INTRODUCTION TO DISTRIBUTED DATA PROCESSING

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Disadvantages of Centralized Database (Need for DDBMS)

1. **Single Point of Failure** – Entire system fails if the central server crashes.
2. **Scalability Issues** – Struggles to handle large-scale data and user growth.
3. **Performance Bottleneck** – High traffic can overload the central server.
4. **Slow Access for Remote Users** – Increases latency for distant users.
5. **Limited Availability** – Downtime affects all users and operations.
6. **High Network Load** – Constant data requests slow down performance.
7. **Security Risks** – A single breach can compromise all data.
8. **Data Locality Issues** – Users far from the central server experience delays.
9. **Maintenance Challenges** – Requires strong infrastructure and backup solutions.
10. **Costly Upgrades** – Expanding storage and processing power is expensive.



Feature	Centralized DBMS	Distributed DBMS
Data Storage	Stored at a single location	Stored across multiple sites
Failure Impact	Single point of failure	Fault-tolerant, multiple sites
Scalability	Limited expansion	Easily scalable
Performance	Slower with high traffic	Faster due to load distribution
Access Speed	Slow for remote users	Faster, data closer to users
Maintenance	Easier to manage	Complex due to multiple sites
Cost	Lower setup cost	Higher infrastructure cost
Data Consistency	Easier to maintain	Difficult due to replication
Security	Centralized control	More challenging across sites
Query Processing	Simple, local execution	Distributed, requires optimization

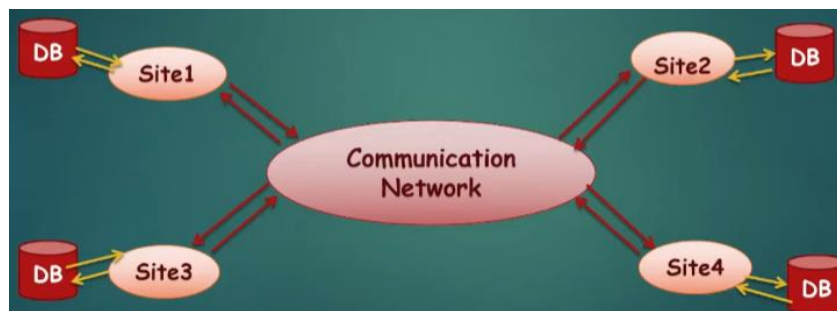
Distributed Data & Database

Distributed Data refers to data that is not stored in a single location but is instead spread across multiple physical or logical locations, such as different servers, cloud storage, or databases. The key idea is that data is stored closer to the point of use, improving accessibility, availability, and performance.

- Stored across multiple sites or servers.
- Managed independently at each site but still accessible globally.
- Can be **replicated** (same data stored in multiple locations) or **fragmented** (different parts of the data stored at different locations).
- Requires a mechanism to ensure **data consistency and synchronization** across all sites.

Distributed database is collection of multiple logically interrelated databases distributed over a computer network (as if they are single database).

- **Database System** – Manages structured data, ensuring storage, retrieval, integrity, and security.
- **Computer Networks** – Enables communication between distributed database nodes across different locations.



Examples of Distributed Data:

- **Banking Systems:** Customer account data is stored at different branch locations but can be accessed globally.
- **Cloud Storage Services:** Google Drive, Dropbox, and OneDrive store user data across multiple data centres.
- **E-commerce Platforms:** Amazon stores inventory data in different warehouses, ensuring fast access to stock information.

Distributed Database Management System (DDBMS)

A **Distributed Database Management System (DDBMS)** is a software system that manages a distributed database, ensuring that the data appears as a single database to users while being stored across multiple locations.

1. **Data Distribution** – Uses fragmentation, replication, and allocation to distribute data.
2. **Query Processing** – Splits queries, optimizes execution, and minimizes overhead.
3. **Transaction Management** – Maintains ACID properties across sites.
4. **Concurrency Control** – Prevents conflicts using locking and timestamps.
5. **Replication & Synchronization** – Ensures consistency across replicated data.
6. **Failure Recovery** – Maintains consistency and system continuity after failures.
7. **Security** – Uses authentication, authorization, and encryption for protection.
8. **Distributed Transactions** – Uses protocols like 2-phase commit to ensure atomicity.

Implicit Assumptions in a Distributed Database System (DDBMS):

1. **Data Distribution** – Data is stored across multiple sites, each functioning as a single logical processor.
2. **Network Connectivity** – Sites communicate via a computer network.
3. **Database Nature** – The system manages structured data, not just a collection of files.
4. **User Access Patterns** – Data placement and query execution depend on user access behavior.
5. **DDBMS Structure** – It is a collection of DBMSs managing distributed data, not a remote file system.

Applications of DDBMS:

1. **Banking**: Manages accounts and transactions across branches.
2. **Telecom**: Handles customer data and call records globally.
3. **E-commerce**: Stores product data and orders across locations.
4. **Cloud Computing**: Distributes user data for scalability.
5. **Healthcare**: Manages patient records across hospitals.
6. **Supply Chain**: Tracks inventory and shipments globally.
7. **Social media**: Stores user data across multiple servers.

Advantages:

1. **Reliability**: Ensures availability even during failures.
2. **Scalability**: Expands easily with more sites.
3. **Locality**: Data stored closer to users.
4. **Performance**: Reduces bottlenecks via distribution.
5. **Control**: Centralized management with distributed access.
6. **Security**: Uses encryption and partitioning.

Disadvantages:

1. **Complexity**: Difficult to design and manage.
2. **Cost**: Expensive to implement and maintain.
3. **Overhead**: High communication and synchronization costs.
4. **Consistency**: Challenging to maintain uniform data.
5. **Security Risks**: Harder to secure multiple sites.

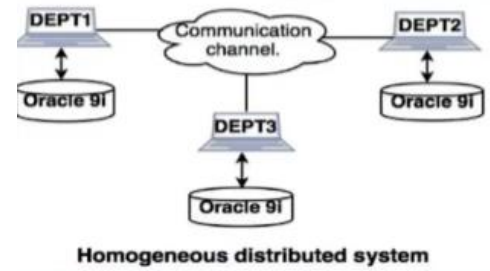
Date's 12 Rules for DDBMS

1. **Local Autonomy** – Each site can function independently.
2. **No Reliance on a Central Site** – No single point of control or failure.
3. **Continuous Operation** – System should remain operational even if some sites fail.
4. **Location Independence** – Users can access data without knowing its physical location.
5. **Fragmentation Independence** – Data can be fragmented across sites without affecting users.
6. **Replication Independence** – Users should not be concerned with whether data is replicated.
7. **Distributed Query Processing** – Queries should be optimized across multiple sites.
8. **Distributed Transaction Management** – ACID properties should be maintained across sites.
9. **Hardware Independence** – Should work across different hardware platforms.
10. **Operating System Independence** – Should function across multiple OS environments.
11. **Network Independence** – Should operate over various network types.
12. **DBMS Independence** – Should support different database systems at different sites.

Types of DDBMS

1. **Homogeneous Distributed Database System:** All the sites in a homogeneous DDBMS use the same DBMS software and hardware. **Example:** Multiple instances of Oracle DBMS across different locations.

- Consistent data model and schema across all sites.
- Easier to maintain as the software is uniform.
- Simplified data management and administration.
- Seamless integration between different sites.
- Example:** some changes made in one of the 3 departments, will be reflected and updated in other 2.



f. **Types of HDBBMS:**

- Autonomous** – Each site has its own database and transaction management but follows a common schema and DBMS.
- Non-Autonomous** – A central system controls all sites, enforcing uniform transaction management and coordination.

Site Autonomy (Local Autonomy)

Site autonomy (or local autonomy) means that each site in a DDBMS can operate independently without requiring constant coordination from a central authority.

Key Aspects:

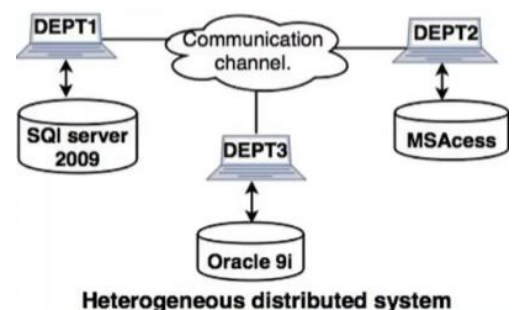
- Independent Control** – Each site manages its own database, transactions, and recovery.
- Local Transaction Execution** – Sites can process local queries without external dependency.
- Failure Isolation** – If one site fails, others continue functioning.
- Decentralized Administration** – No single site has complete control over the entire system.
- Flexibility** – Sites can adopt their own security and access policies.

Types:

- Full Autonomy:** Sites operate entirely independently, with minimal synchronization.
- Partial Autonomy:** Some degree of central coordination exists, but sites have local control over data.

2. **Heterogeneous Distributed Database System:** The sites in this system may use different DBMS software or hardware, which may lead to variations in data models and schemas. **Example:** Integration of various DBMS like MySQL and MongoDB at different sites.

- Different types of DBMS are used at different sites (e.g., Oracle, MySQL, SQL Server).
- Requires middleware to enable interoperability between different systems.
- Flexibility in selecting the best DBMS for each site.
- Ability to integrate different types of systems.

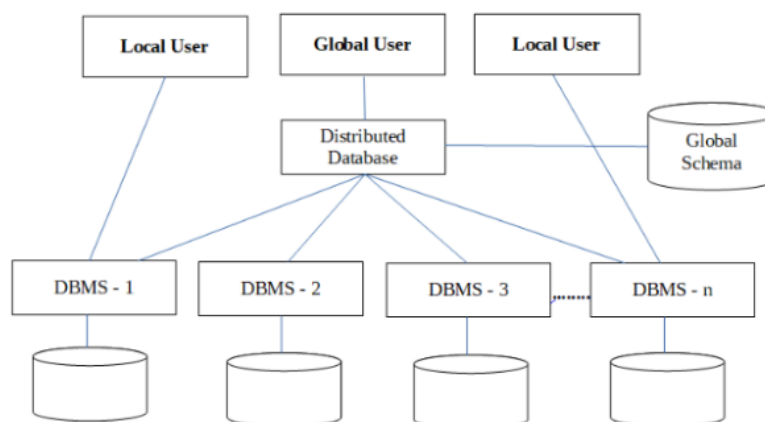


Feature	Homogeneous DDBMS	Heterogeneous DDBMS
Definition	All sites use the same DBMS	Sites use different DBMSs
Schema	Common global schema	Different schemas across sites
Interoperability	High, easy communication	Low, requires translation
Autonomy	Usually non-autonomous	Can be autonomous
Complexity	Lower, easier to manage	Higher, needs middleware
Query Processing	Simple and uniform	Complex, requires integration
Transaction Management	Easier to maintain ACID	More challenging to ensure ACID
Examples	Multiple Oracle DBs across sites	Oracle at one site, MySQL at another

3. **Fragmented Distributed Database System:** The database is divided into fragments (subsets of data), which can be distributed across different sites. **Example:** Customer database fragmented by region, where each regional office stores its own data.
 - a. Data is divided based on rows (horizontal fragmentation), columns (vertical fragmentation), or a combination of both.
 - b. Each site holds only a part of the data, and the database is recreated by collecting fragments.
 - c. Efficient data access and reduced load at each site.
 - d. Localized queries to only relevant data.
4. **Replicated Distributed Database System:** Copies (replicas) of the same data are stored at multiple sites, ensuring data availability and fault tolerance. **Example:** Online services like Amazon replicate their product catalog across multiple servers.
 - a. Data replication improves the reliability and availability of data.
 - b. Updates to data need to be synchronized across all replicas.
 - c. High availability of data even in the event of site failure.
 - d. Faster read access to data due to replication.
5. **Hybrid Distributed Database System:** Combines both fragmented and replicated distributed database systems. Some data is fragmented and distributed, while other data is replicated across sites. **Example:** A global e-commerce platform using fragmented product catalogs and replicated customer data.
 - a. Flexibility in data management, allowing both distribution and replication.
 - b. Some parts of the database are fragmented based on the workload, while others are replicated for redundancy.
 - c. Optimized performance by balancing between fragmentation and replication.
 - d. Scalable and fault-tolerant system.

Components of a DDBMS

A **DDBMS** consists of several components that work together to provide a unified view of data spread across different locations. These components ensure that data is accessible, consistent, and manageable in a distributed environment.



1. **Database Sites:** These are the physical locations where the database is stored. Each site may contain a fragment of the data, or the complete database, depending on the type of distribution (fragmentation or replication). **Types:**
 - **Master Site:** Primary location responsible for central coordination, may hold the complete database.
 - **Slave Sites:** Secondary locations that may store fragments or replicas.
2. **Fragmentation, Replication, and Allocation Control:** This component handles the distribution of data across various sites. It decides how data is fragmented (row/column), replicated, and allocated across different nodes.
3. **Query Processor:** The query processor breaks down user queries and distributes them to appropriate sites for execution. It optimizes query execution by considering where data is stored. **Components:**
 - **Parser:** Checks the syntax and semantics of the query.
 - **Optimizer:** Determines the most efficient execution plan for queries.
 - **Execution Engine:** Executes the query on the relevant site(s).
4. **Transaction Management:** Responsible for ensuring that distributed transactions maintain **ACID (Atomicity, Consistency, Isolation, Durability)** properties across all sites. It manages coordination and synchronization between various sites during a transaction.
5. **Concurrency Control:** Manages simultaneous data access to ensure consistency when multiple users are accessing and modifying data at different sites. **Techniques:** Locking protocols, timestamp ordering, and multi-version concurrency control (MVCC).
6. **Communication Subsystem:** Facilitates communication between distributed database sites. This includes both message passing and data exchange. It ensures reliable data transmission, handles failures, and manages network protocols.

7. **Recovery Subsystem:** Ensures that the database can recover from failures, including network, hardware, and software failures. It guarantees that data remains consistent and the system can resume operations after a failure.
8. **Directory Service:** Keeps track of the metadata, which includes information about where each piece of data is located (i.e., the location of fragments or replicas). It helps in locating data efficiently and guiding the system in accessing relevant data.

Questions

1. Storing a separate copy of the database at multiple locations is which of the following?

- a. Data Replication
- b. Horizontal Partitioning
- c. Vertical Partitioning
- d. None of the above

✓ **Answer:** a. Data Replication

2. A distributed database is a collection of data which belong _____ to the same system but are spread over the _____ of the network.

- a. Logically, sites
- b. Physically, sites
- c. Database, DBMS
- d. None of the above

✓ **Answer:** a. Logically, sites

3. Which of the following is/are the main goals of a distributed database?

- a. Interconnection of database
- b. Incremental growth
- c. Reduced communication overhead
- d. All of the above

✓ **Answer:** d. All of the above

4. Which of the following is a distributed database?

- a. A single logical database that is spread to multiple locations and is interconnected by a network
- b. A loose collection of a file that is spread to multiple locations and is interconnected by a network
- c. A single logical database that is limited to one location
- d. None

✓ **Answer:** a. A single logical database that is spread to multiple locations and is interconnected by a network

5. Which of the following strategies is used by a distributed database?

- a. Totally centralized at one location and accessed by many sites
- b. Totally or partially at one location and distributed at many sites
- c. Partitioned into segments at different sites
- d. All of the above

✓ **Answer:** d. All of the above

6. An autonomous homogeneous environment is which of the following?

- a. The same DBMS is at each node and each DBMS works independently.
- b. The same DBMS is at each node and a central DBMS coordinates database access.
- c. A different DBMS is at each node and each DBMS works independently.
- d. A different DBMS is at each node and a central DBMS coordinates database access.

☒ **Answer:** a. The same DBMS is at each node and each DBMS works independently.

7. A heterogeneous distributed database is which of the following?

- a. The same DBMS is used at each location and data are not distributed across all nodes.
- b. The same DBMS is used at each location and data are distributed across all nodes.
- c. A different DBMS is used at each location and data are not distributed across all nodes.
- d. A different DBMS is used at each location and data are distributed across all nodes.

☒ **Answer:** d. A different DBMS is used at each location and data are distributed across all nodes.

8. Which of the following is not one of the stages in the evolution of distributed DBMS?

- a. Unit of work
- b. Remote unit of work
- c. Distributed unit of work
- d. Distributed request

☒ **Answer:** a. Unit of work

9. Databases that are stored on computers at multiple locations and are not interconnected by a network are known as distributed databases.

- a. True
- b. False

☒ **Answer:** b. False

10. Which of the following is the advantage of a distributed database over a centralized database?

- a. Module wise growth
- b. Simplicity and ease
- c. Slow Response
- d. None of the above

☒ **Answer:** a. Module wise growth

11. Which DDBMS provides better _____ over centralized DBMS?

- a. Decentralization
- b. Tuning
- c. Security
- d. Transparency

☒ **Answer:** a. Decentralization

12. Which of the following features of DDBMS?

- a. Distribution
- b. Logical correlation
- c. Distribution & logical correlation
- d. Disjointness

☒ **Answer:** c. Distribution & logical correlation

13. What is site autonomy?

- a. Each server participating in a distributed database is administered independently from all other databases.
- b. Nodes of the system cannot mirror the logical organization of companies or groups that need to maintain independence.
- c. Local administrators cannot control corresponding local data. Therefore, each database administrator's domain of responsibility is smaller and more manageable.
- d. Administrators cannot recover from isolated system failures independently from other nodes in the system.

✓ **Answer:** a. Each server participating in a distributed database is administered independently from all other databases.

14. A set of databases in a distributed system that can appear to users as a single data source is known as

- a. Distributed database
- b. Distributed processing
- c. Both A & B
- d. None of the above

✓ **Answer:** c. Both A & B